MILITARY SPECIFICATION
FOR
INDICATOR, HUMIDITY, PLUG

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Indicator, humidity, plug used on containers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-8835 --- Indicator, Humidity, Card, Chemically Treated
MIL-I-26860 -- Indicator, Humidity, Plug, Color Change
MIL-P-116 ---- Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging and Marking for Shipments of Inert Ammunition Components

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DRAWINGS

ORDNANCE CORPS

8791666 ---- Indicator, Humidity, Plug
8791668 ---- Gasket
8791669 ---- Window Observation
8791670 ---- Gasket
8791671 ---- Nut, Plain, Hexagon
8791672 ---- Plate, Indicator
8791673 ---- Ring Externally Threaded
8791682 ---- Housing, Card, Humidity Indicator
8808362 ---- Packing and Marking for Storage and Shipments
8875803 ---- Indicator, Humidity Card
8883538 ---- Bag, Packing for Repair Parts
ET-8791666 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The indicator humidity plug shall comply with all requirements specified on the drawing (dwg) 8791666, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

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3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I

Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table II

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

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4.3.2.1.1 Gasket (see dwg 8791668) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Outside diameter, min. (1.185)	Gage
102.	Thickness (.073)	Gage
103.	Inside diameter, max. (1.045)	Gage
104.	Material (see 3.1)	Visual
Minor		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.2 Window, observation (see dwg 8791669) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Outside diameter, min. (1.167)	*Measure
102.	Thickness (.156)	*Measure
103.	Material (see 3.1)	Visual

*Measure with suitable scale.

Minor		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.3 Gasket (see dwg 8791670) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Outside diameter, min. (1.670)	Gage
102.	Thickness (1.06)	Gage
103.	Inside diameter, max. (1.472)	Gage
104.	Material (see 3.1)	Visual
105.	No cracks or holes	Visual

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4.3.2.1.4 Nut, plain, hexagon (see dwg 8791671) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Distance across flats (1.625)	Gage
102.	Pitch diameter of thread, max.	Gage
103.	Minor diameter of thread, max.	Gage
104.	Thickness (3.25)	Gage
105.	Material (see 3.1)	Visual
106.	Objective evidence (see 6.2) for finish in accordance with dwg 8791671	Visual
Minor		
201.	Surface finish	Visual
202.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.5 Plate, indicator (see dwg 8791672) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Outside diameter, max. (1.22)	*Measure
102.	Diameter of holes, min. (.250)	Gage
103.	Total thickness (.100)	*Measure
104.	Thickness of material (.032)	Gage
105.	Material (see 3.1)	Visual
106.	Objective evidence (see 6.2) for finish in accordance with dwg 8791672	Visual
	*Measure with suitable scale.	
Minor		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.6 Ring, externally threaded (see dwg 8791673) covering a detail of dwg 8791666.

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Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Inside diameter, min. (1.046)	Gage
102.	Width of slots (.060)	Gage
103.	Depth of slots, min. (.052)	Gage
104.	Distance from left face to groove (.093)	Gage
105.	Total length (.343)	Gage
106.	Width of groove (.046)	Gage
107.	Pitch diameter of thread	Gage
108.	Major diameter of thread	Gage
109.	Material (see 3.1)	Visual
110.	Objective evidence (see 6.2) for finish in accordance with dwg 8791673	Visual
Minor		
201.	Surface finish	Visual
202.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.7 Housing, card, humidity indicator (see dwg 8791682) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Outside diameter (1.84)	*Measure
102.	Inside diameter (.99)	Gage
103.	Width of slots (.053)	Gage
104.	Depth of slots (.052)	Gage
105.	Diameter of chamfer (1.211)	*Measure
106.	Distance from left fact to groove (.233)	Gage
107.	Distance from bottom of inside thread to flange shoulder (.125)	Gage
108.	Distance from left face to bottom of cavity (.197)	Gage

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109. Thickness of flange (.187).....Gage
110. Pitch diameter of outside thread.....Gage
111. Major diameter of outside thread.....Gage
112. Pitch diameter of inside thread.....Gage
113. Minor diameter of inside thread.....Gage
114. Chamfer on threads present.....Visual
115. Total length (.71).....*Measure
116. Small diameter of groove in flange (1.474).....Gage
117. Width of groove in flange (.112).....Gage
118. Radius present.....Visual
119. Depth of groove in flange (.070).....Gage
120. Width of outside groove (.03).....Gage
121. Width of inside groove (.075).....Gage
122. 63 finish.....Visual
123. Diameter of outside groove (1.36).....Gage
124. Small diameter of inside groove (1.044).....Gage
125. Depth of inside groove (.050).....Gage
126. Material (see 3.1).....Visual
127. Objective evidence (see 6.2) for finish in accordance with dwg 8971682.....Visual

*Measure with suitable scale.

Minor
201. Surface finish except as noted.....Visual
202. Stamping.....Visual
203. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.8 Indicator, humidity, card (see dwg 8875803) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major
101. Outside diameter (1.20).....Gage
102. Objective evidence (see 6.2) for card in accordance with MIL-I-8835 as per drawing.....Visual
103. Marking.....Visual

Minor
201. Workmanship (see 3.3).....Visual/Manual

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4.3.2.1.9 Indicator, humidity, plug (see dwg 9271666).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	No parts missing, properly assembled	Visual
Minor		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.10 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Preservation, packaging and marking in accordance with MIL-STD-1169	Visual/Manual
Minor: None defined.		

4.3.2.1.11 Bag, packing for repair parts (see dwg 8883538) covering part number 8883538-3 and 5.2.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Material (see 3.1)	Visual
102.	Length before packing, min.	*Measure
103.	Width before packing, min.	*Measure
104.	Properly packed	Visual/Manual
105.	Proper heat seal	Visual/Manual
106.	No crack or hole in bag	Visual
107.	Objective evidence (see 6.2) for Heat Seal Test in accordance with MIL-P-116	Visual

*Measure with suitable scale.

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Minor
201. Marking ..... Visual
202. Workmanship (see 3.3) ..... Visual/Manual

4.3.2.1.12 Packing and marking for storage and shipments (see dwg 8808362 and 5.2).

Categories	Characteristics	Method of Inspection

Critical: None defined

Major
101. Gross load weight, max. ..... *Measure
102. Contents tight in package ..... Manual
103. Correctly packed ..... Visual/Manual
104. Material (see 3.1) ..... Visual

*Determine weight with suitable scale.

Minor
201. Marking ..... Visual
202. Workmanship (see 3.3) ..... Visual/Manual

4.3.3 Testing. All tests shall be performed. Tests performed in the order specified.

Test	Test Classification
a. Acceptance tests	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Acceptance tests (see dwg 9271666). The indicator, humidity, plugs shall be tested in accordance with 4.4.1. If the indicator, humidity, plugs fail to comply with the requirements of MIL-I-26860, the lot of indicator, humidity, plugs shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL's). EL's for assemblies are listed on the Equipment Tabulation, ET-8791666. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Acceptance tests (see 4.3.3.1). The indicator, humidity, plugs shall be tested as specified for Acceptance Tests in MIL-I-26860.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The indicator, humidity, plug shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The indicator, humidity, plug shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

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5.1.2 Packing.

5.1.2.1 Levels A and C. The indicator, humidity, plug shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The indicator, humidity, plug shall be packaged in accordance with dwg 8883538 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The indicator, humidity, plug shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The indicator, humidity, plug shall be packed in accordance with dwg 8883538 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 8883538 and 8808362 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.

b. Production or Repair Parts.

c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on DI-E-1118 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SMUPA-QA-N-I.

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Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 6685-A466

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DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SARPA-DA-H-Q
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-I-48191(MU) Indicator Humidity Plug

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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